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EDUCATION OF MSC AND PHD STUDENTS IN FLUID POWER AND MECHATRONICS AT DTU

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ABSTRACT

This paper deals with education of MSc and PhD students in the engineering areas fluid power and mechatronics. The Technical University of Denmark, DTU has developed a novel education structure and programs for MSc and PhD students adapted to the change and development of technologies. The MSc students may decide to specialise in one of the four classical engineering fields: Mechanical, Electrical, Civil or Chemical Engineering and/or in one of 20 specific profiles of engineering. One of these profiles is *Engineering Design and Product Development* that gives possibility for specialisation in fluid power system and mechatronics design. Furthermore, the paper describes how synthesis, design and self-learning competency have a very high priority in the teaching and training in industrial enterprises. Projects cover mathematical modelling, simulation, experiments, implementation and prototyping.

KEY WORDS

Engineering education, fluid power, mechatronics, control and engineering design, product development.

1. INTRODUCTION

Product life cycles have decreased dramatically driven by rapid technology changes, increasing global public demands, requirements and regulations for protection of the environment. As a consequence the enterprises have to cope with an increasing competition on the world market. There is always an ongoing desire to develop and design new products and systems to improve the performance specifications, in particular focus is on productivity and efficiency of present functionality and control of process operations. Today, the demands, requirements and regulations are more complex because they include customers demands, requirements for extensions of functionality, capability, robustness, reliability etc., and many public regulations such as EU-regulations, US-regulations, Australian-regulations etc.

Today's information technology offers both software and hardware for engineering solutions which could satisfy such demands, requirements and regulations. On the hardware side, actuators, sensors, fast and powerful computers, microcomputers, digital signal processors and transputers are available for control, monitoring and supervision of systems. On the software side, modelling and simulation software and real-time software are available for design and implementation of advanced operator-assisted control systems and *intelligent control systems* using simple fuzzy logic and expert systems to achieve autonomous control systems.

In response to the significant challenge described above the Department of Control and Engineering Design, DTU has developed a novel MSc Program with several new courses for students who want the profile *Engineering Design and Product Development*. Within the profile the students can specialise in fluid power system and/or mechatronics design. The department do also educate several PhD students in these areas. The software packages like Matlab, Simulink, Vissim, Bathfp Mechanica and Proengineer are important tools.

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In both manufacturing and non-manufacturing industries there are significant needs to improve performance and cut down of costs and delivery times. The trend of engineering solutions is design of intelligent products, machines, machinery, robots and integrated production systems to improve the performance, to optimise productivity and efficiency, to offer a intelligent human-based man-machine and man-model interface, and to minimise resource consumption and pollution of the environment. Summarised, the successful companies on the world market are those with the ability to apply technologies in order to deliver products and systems with the targets to:

- increase performance, in particular functionality
- increase robustness, reliability and safety
- increase benefits to customers
- minimise product development and design cycles (faster prototyping and faster delivery)
- obtain lower costs (including operation cost per year)
- improve recycling

The ability to apply the best technologies for those targets are the companies who have the best qualified design engineers and researchers educated to work in teams who are able to manage the new technologies and who are specialised in an integrated product development approach to control and engineering design, in particular mechatronics.

2. DTU'S NEW FACULTY STRUCTURE

DTU was recently reorganised in five faculty committees, each responsible for their own general discipline:

- Machine Technology, Energy and Production
- Structural Engineering, Environmental Engineering and Planning
- Chemistry, Biotechnology and Food Sciences
- Communications, Computer Sciences and Mathematics
- Electrical Engineering and Physics

These committees advise the *Konsistorium*, the supreme body of DTU, on all academic matters. The Department of Control and Engineering Design belong to the first mentioned faculty committee.

Each program, Bachelor's, Master's and PhD's also has a Program Council consisting of an equal number of teachers (professors or associate professors) and students. The three programs councils at DTU deal with matters concerning the various major subjects: they approve teaching plans, assign teaching resources and draw up curriculum proposals. The council chairmen are also authorized by the Rector of DTU to run their programs on a day-to-day basis. The population of students is approximately the following:

- Bachelor's Program, 3 1/2 years with 2,000 students,
- Master's Program, 5 years with 4.900 students,
- PhD's Program, 3 years with 650 PhD students.

All teaching in the Master's Program is and must always be based on research. This a fundamental principle at DTU aimed at ensuring high standards in the Program. The DTU Staff numbering almost 1,900 full-time.

3. MSC PROGRAMS AND PROFILES

About 4,900 students are enrolled in the Master's Program, which takes five years (10 semesters) to complete. Today approximately 20% of the students enrolled are women. Students going for a Master's Degree in Engineering from DTU must enroll in one of eight preliminary four-semester programs:

- Mechanical Engineering
- Energy
- Informatics
- Electrical Engineering
- Applied Physics
- Chemical Engineering
- Civil- & Construction Engineering
- Environmental Engineering

Each program consist of compulsory courses in Mathematics and Physics and basic courses in the specific engineering field chosen by the student. The program also includes courses dealing with work environments, social science subjects, and relationship between technology and environment. Even though these courses are compulsory, it is still possible for students to put together their own curriculum to a certain extent.

During the first two years of studies, students are mainly instructed in classes and work in groups. Later on, lectures and laboratory courses are the most common form for instruction in the Master's Program. After the first two years, it is very much up to the student to plan his/her own curriculum according to inclination, interest and prerequisites with respect to both length and study and choice of future occupation. About 700 courses are offered, and students are expected to pass five courses per semester in order to finish their studies in five years. The students have to pass about 50 courses during their studies.

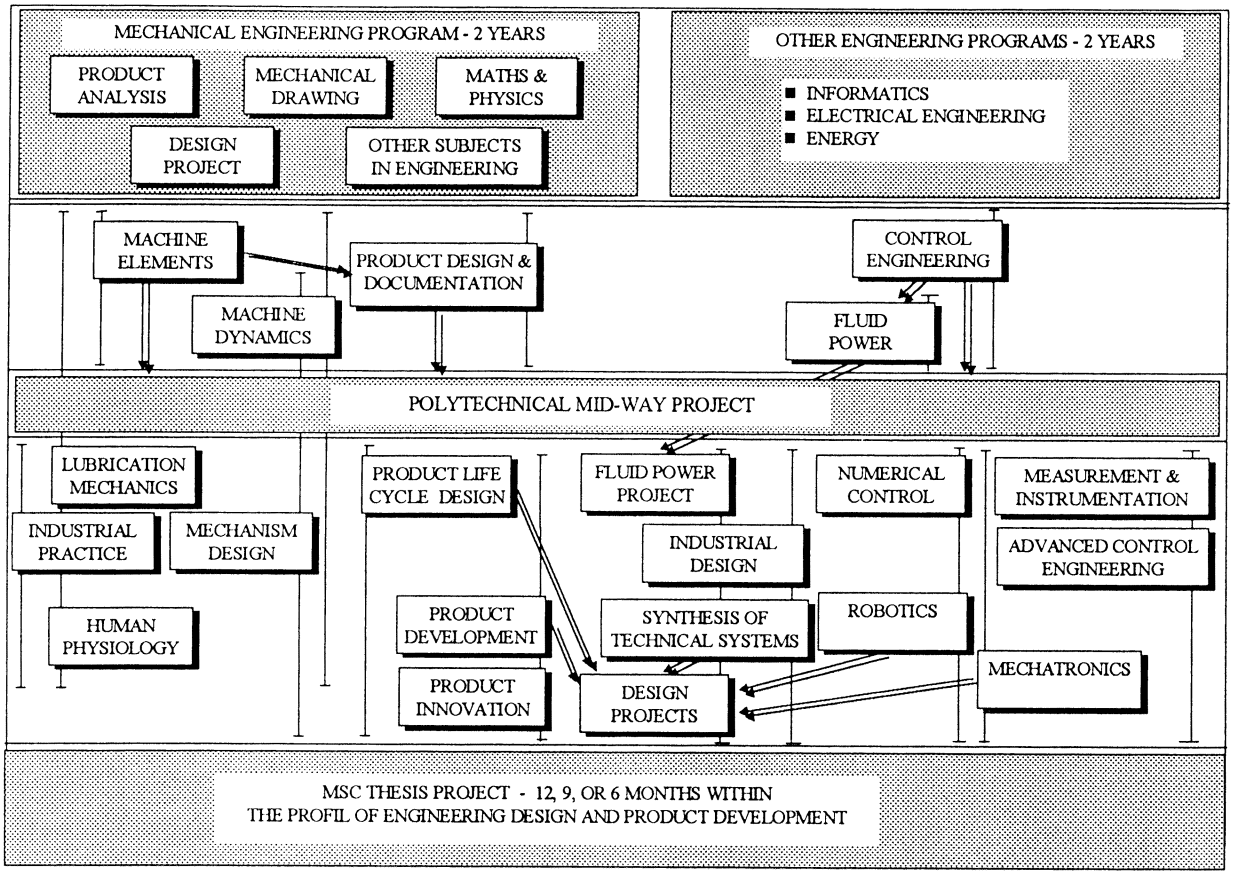


Figure 1 Schematic of the MSc Program for the profile of *Engineering Design and Product Development.*

Halfway through their studies, students must do substantial body of work in a semester in the form of a *Polytechnic Mid-way Project.* The project has a study load as three normal lecture courses. Small groups of students usually work together on one project, which may be performed in cooperation with a firm. A course of study ends with a Master's Thesis Project which takes between five and twelve months to complete.

Depending on the students choice of courses and planning of curriculum, they may decide to specialise in one of the four classic engineering fields: Mechanical, Electrical, Civil or Chemical Engineering and/or in one of the 20 novel specific profiles of engineering:

- Applied Mechanics
- Biotechnology
- Civil Engineering Structures
- Computer Systems
- Microelectronics
- Energy Technology
- Pure and Applied Chemistry
- Ocean Engineering
- Manufacturing
- Informatics
- Communication
- ***Engineering Design and Product Development***
- Materials Science and Engineering
- Environmental Engineering
- Planning and Technology Management
- ***Control Engineering***
- Applied Mechanics
- Chemical Engineering
- Industrial Electrical Engineering
- The Built Environment

The Department of Control and Engineering Design, DTU has developed and is responsible for the novel MSc Program with several new courses for students who want the qualification profile *Engineering Design and Product Development, EDAPD.* Figure 1 shows a schematic of this program. Further, the department is in the consortium of departments who have developed the profile of engineering *Control Engineering, CE.*

Synthesis, design and development of the students self-learning competency have a high priority in the teaching and training of the EDAPD profile and the courses and

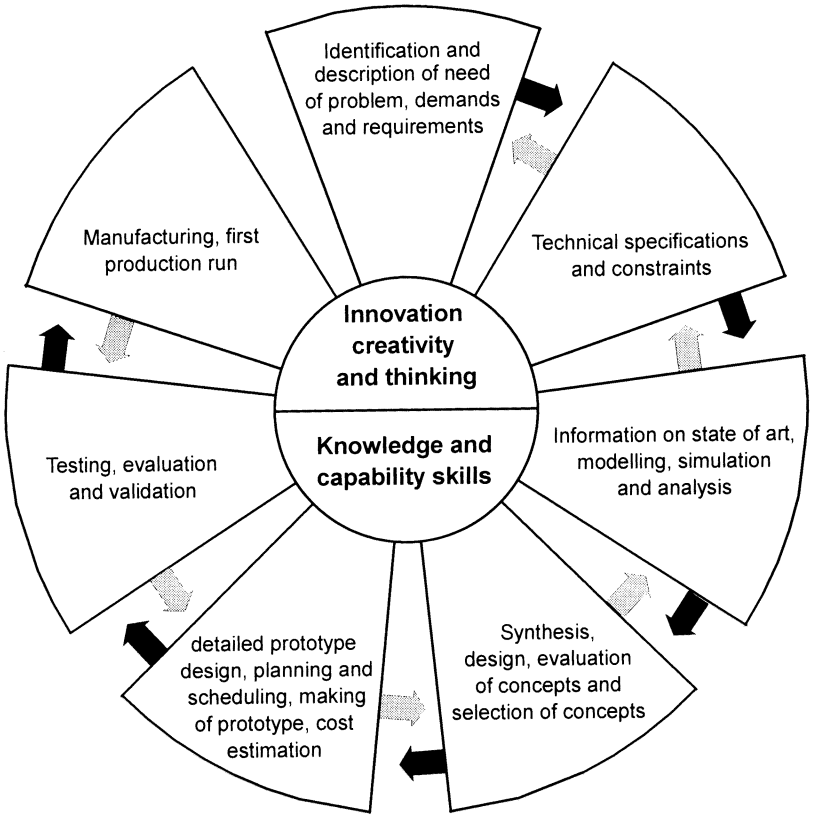


Figure 2 Schematic of engineering design approach.

projects which the department are responsible for in the EC profile. The students have to carry out several projects, some of them in cooperation with industrial enterprises. The focus on self-learning competency has the target to develop the students ability to effective life-long active learning. The program is flexible in such a way that profile students can specialise on one of a few specific key topics. Two of the key topics are fluid power system design and mechatronics design. A third novel key topic is *integrated product development*. As a novel subject the department offers specialisation in *pure water hydraulic control engineering* in cooperation with Danfoss A/S.

The new students who are going for a Master's Degree in Engineering are enrolled the Master's Program, which takes five years (10 semesters) to complete. Students must enroll in one of the eight preliminary four-semester programs mentioned above. Four of the preliminary programs

- Mechanical Engineering
- Energy
- Informatics
- *Electrical Engineering*

are a basis for specialisation for the EDAPD profile.

The Mechanical Engineering Program has two project courses: *Product Analysis* and *Design Project*. The objective of the first course is to teach the students to obtain a fundamental understanding of machine systems functionality, working principles, construction and manufacturing. Focus is on modelling and description of technical components and systems. The students are organised in group of four and taught to understand that a complete system should have good individual components, and however, good components of itself does not guarantee a system with good performance because the overall system performance depend of the static and dynamic interaction between the components or the sub-systems. Stability and control is introduced in that respect. The groups describe the results of the project work in a report and an oral presentation followed by discussion and the evaluation.

The objective of the other course Design Project is to teach and give the students experience in engineering design. Focus is on the nature of engineering design work, synthesis and methodologies, development of the students ability to understand technical products and systems, to concept-thinking, to create and contribute to innovation. Project subjects are within: design, machine elements and components, synthesis of technical systems, mechatronics, robotics, life-cycle engineering design and

product development. Concept thinking, modelling and simulation are major subjects in the project work.

On the 4-6 semester the students take courses in control engineering, machine elements, machine dynamics, fluid power and product design.

On the 5-6 semester the students have to carry out the *Polytechnic Mid-way Project*. Small groups of students usually work together on one project, which may often be in cooperation with a company.

From 6 semester starts the profile of engineering specialisation within the topics shown in the schematic in Figure 1. The study ends with the Master's Thesis Project which takes between six and twelve months to complete. Very often the project is linked to ongoing research and/or product development in cooperation with a company.

Mechatonics, fluid power systems design, engineering design of products and systems, life-cycle engineering design, robotics, integrated productions systems and integrated product development are topics for the recently and future projects. Projects on *pure water hydraulic systems* are undertaken in cooperation with Danfoss A/S. Figure 3 shows the today available Nessie family of water hydraulic components for such projects.

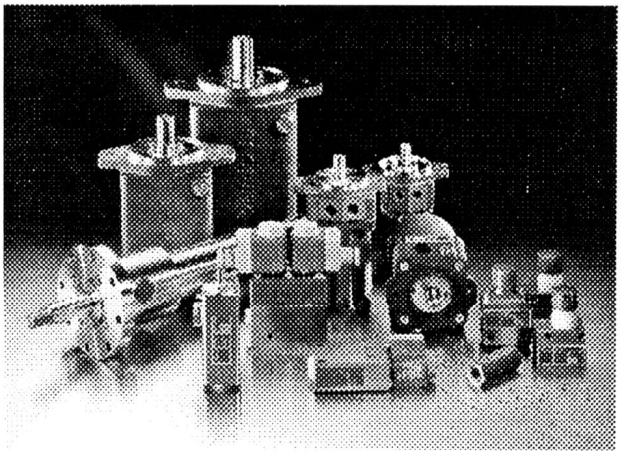


Figure 3 Pure water hydraulic components (Danfoss).

The approach to mechatronics is based on mechatronics to be seen as primary, secondary, and tertiary [1, 2, 3]. The *primary level* encompasses devices which integrate electronic signalling with mechanical action at a basic local level. An electrically-controlled digital fluid valve is an example. The *secondary level* implies a local stand-alone but complete system which uses integration of electronics with mechanical action. A disc player is in this category.

The *tertiary level* implies a more multi-funtional and sophisticated system of integrated software, electronics and mechanical movements with advantaged state sensing, control and monitoring. The key features of tertiary mechatronics are *System Conception and Design, On-line Intelligent Control*, and *Powered Actuation*. Figure 4 shows schematically the ingredients for tertiary mechatronics. A modern industrial robot with some sensing and intelligence is an example of tertiary level mechatronics. The interdisciplinary enabling technologies and techniques can be fitted into one of these three strategic groups.

Design of intelligent control systems, robots, machines, actuators and components implies awareness of all of the techniques and technologies available to tertiary mechatronics; the ability to properly appreciate the application for which a mechatronics solution is to be sought; conception of strategies and then system components (hardware and software); and detailed specifications such that the mechatronics system can be realised and installed.

Control implies continual on-line state sensing; management of that data and desired state data in an intelligent on-line way; decision-making on control or corrective action, possibly using forecasting techniques. Actuation implies that after all sensing and decision making is done, muscle is needed to perform the decided actions. Without Actuation, we have a head but no torso - we can see and decide what should be done, but can't actually do it.

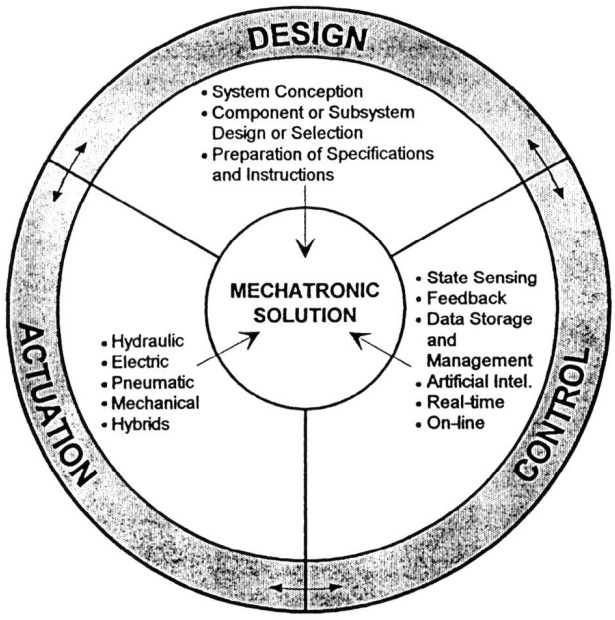


Figure 3 Ingredients for tertiary mechatronics [1, 2, 3].

The powered actuation technologies are essential to physical mechatronics. Fluid Power, in both its hydraulic and pneumatic technologies, is placed squarely in Mechatronics.

As shown in Figure 5 and 6 the teaching and training of the student is based on developing their self-learning competency in active learning environment.

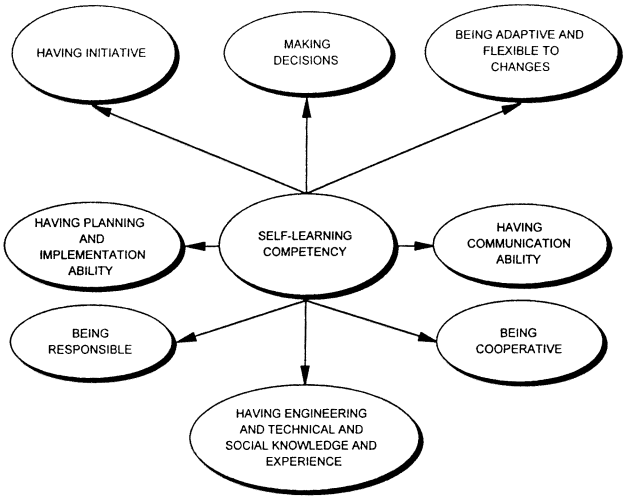


Figure 5 Self-learning Competency.

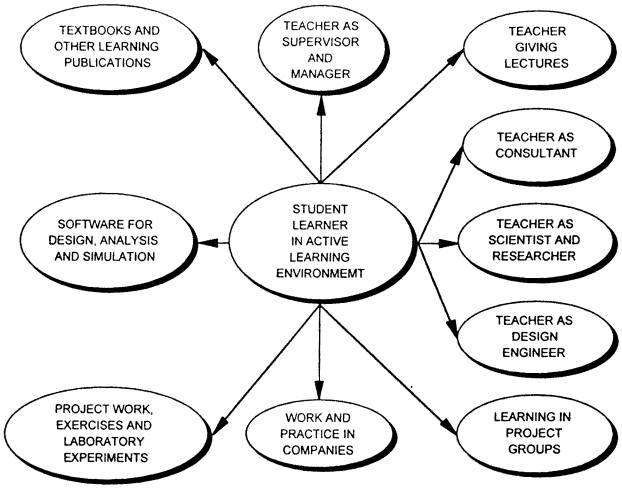


Figure 6 Student in active learning environment.

DTU is associated with technical universities and faculties all over the world in form of student and staff exchange programs, research programs and personal relations among staff members. European Union (EU) Exchange Programs (Erasmus, Tempus, Socrates etc.) have been establish in the recent years. Today about 20% of the students graduating from DTU have spent some time abroad as part of their studies. The number of exchange students are increasing.

4. PHD PROGRAM AND PROFILES

The PhD Program is a 3-year full-time research program taken by a PhD Student under guidance of a responsible supervisor who is Professor or Associate Professor. The requirement to start as a PhD Student is a Master's Degree from a university or an equivalent institution of higher education in Denmark or an equivalent degree from abroad. The Government has laid down the overall framework of the PhD Program to ensure a general measure of quality. It is, however left to the 12 Danish universities authorized to award PhD degrees to design their own specific programs to match their own special needs and traditions at the department level. Consequently, each university has its individually PhD program divided into a number of programs of study. Each study has its own particular course of study and requirements for enrollment.

DTU has seven PhD programs of study. Each program has its own polices and requirements, which are described in special program guides. Each of the seven PhD programs at DTU has a program committee, and each program committee is responsible for the structure and content of its PhD program within the framework of DTU's general policies.

5. CONCLUSION AND PERSPECTIVE

In response to the significant challenge described above the Department of Control and Engineering Design, DTU has developed a novel MSc Program with several new courses for students who want the profile *Engineering Design and Product Development*. Within the profile the students can specialise in fluid power system and/or mechatronics design. A tertiary level of mechatronics approach is used. The department do also educate several PhD students in these areas.

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